

BOWERS FIELD, WA, USAWMO: **727883**Lat: **47.034N**Lon: **120.530W**Elev: **538**StdP: **95.02**Time Zone: **-8.00 (NAP)**Period: **98-19**WBAN: **24220****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-13.8	-10.7	-17.9	0.8	-12.9	-14.8	1.1	-9.3	12.4	5.5	10.9	4.2	0.8	50	0.678

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	36.0	18.3	33.7	17.8	31.5	17.1	19.6	32.8	18.7	31.4	17.8	29.7	3.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	11.1	24.4	13.6	10.4	23.3	12.6	9.7	22.5	57.8	32.4	54.6	31.4	51.7	29.6	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	12.4	11.3	DB	-17.0	39.0	4.2	2.0	-20.0	40.4	-22.4	41.5	-24.8	42.6	-27.8	44.1	(4)
(5)			WB	-17.4	21.4	4.1	1.2	-20.4	22.2	-22.7	22.9	-25.0	23.5	-28.0	24.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-0.9	1.7	5.6	9.3	14.1	17.8	22.3	21.8	16.6	9.7	3.2	-1.4	(6)	
	DBStd	9.04	4.51	3.96	3.61	3.38	4.06	3.90	3.53	3.28	3.65	3.83	4.39	4.40	(7)	
	HDD10.0	1384	339	232	141	53	10	1	0	0	1	49	207	352	(8)	
	HDD18.3	3349	597	465	394	272	141	55	8	8	75	268	454	611	(9)	
	CDD10.0	1396	0	0	4	31	137	236	381	366	199	39	2	0	(10)	
	CDD18.3	320	0	0	0	0	11	41	130	116	23	1	0	0	(11)	
	CDH23.3	4362	0	0	0	21	225	584	1661	1428	426	18	0	0	(12)	
	CDH26.7	2054	0	0	0	3	75	251	855	706	162	3	0	0	(13)	
(14) Wind	WSAvg	4.0	2.0	2.5	3.8	5.0	5.5	6.2	6.2	5.4	4.2	3.2	2.3	1.9	(14)	
(15) Precipitation	PrecAvg	234	32	23	18	17	15	15	8	9	10	15	30	39	(15)	
	PrecMax	359	112	56	86	55	43	43	32	54	46	49	66	107	(16)	
	PrecMin	119	3	0	1	0	0	0	0	0	0	0	0	2	(17)	
	PrecStd	53	22	15	16	11	11	13	8	12	12	12	17	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.8	14.2	20.1	26.3	32.1	36.4	38.7	38.3	34.0	26.1	16.1	10.1	(19)	
		MCWB	7.1	8.2	11.1	13.2	16.4	18.8	18.6	19.1	17.7	14.5	10.8	6.2	(20)	
	2%	DB	7.9	11.4	16.9	22.5	28.7	32.5	36.3	35.6	31.0	21.5	13.3	7.7	(21)	
		MCWB	4.8	6.5	9.2	12.0	15.4	17.2	18.3	18.1	16.4	12.7	9.2	4.9	(22)	
	5%	DB	6.2	9.4	14.1	19.4	25.7	29.3	34.0	33.1	28.1	18.9	11.4	6.0	(23)	
		MCWB	3.7	5.2	7.6	10.6	14.1	16.3	17.9	17.7	15.4	11.3	7.6	3.7	(24)	
	10%	DB	4.3	7.6	12.2	16.6	22.7	26.6	31.8	30.9	25.5	16.9	9.7	4.1	(25)	
		MCWB	2.4	4.2	6.8	9.0	13.0	15.1	17.4	17.0	14.4	10.5	6.4	2.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	9.1	11.8	14.4	17.7	20.1	20.8	20.6	18.9	15.6	11.6	7.0	(27)	
		MCDB	10.3	13.4	18.3	24.3	29.6	32.8	33.7	33.3	31.2	23.7	14.9	9.2	(28)	
	2%	WB	5.2	7.0	9.8	12.5	16.2	18.2	19.8	19.4	17.4	13.7	9.8	5.2	(29)	
		MCDB	7.8	10.7	15.7	21.5	27.1	30.9	33.0	32.7	28.8	19.7	13.3	7.5	(30)	
	5%	WB	3.8	5.7	8.2	11.0	15.0	16.9	18.8	18.4	16.1	12.3	8.1	3.8	(31)	
		MCDB	5.9	8.8	13.4	18.5	24.5	28.0	31.7	30.9	26.2	17.6	10.5	5.8	(32)	
	10%	WB	2.6	4.5	7.1	9.7	13.7	15.6	17.8	17.6	15.1	11.0	6.8	2.6	(33)	
		MCDB	4.2	7.3	11.6	15.9	21.3	25.4	30.0	29.2	23.8	16.0	9.2	4.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.3	9.3	11.7	13.1	13.8	13.9	16.3	16.4	16.0	13.2	9.5	6.5	(35)	
		MCDBR	8.9	12.1	14.8	18.3	19.4	19.3	20.9	21.1	21.0	17.0	11.4	8.6	(36)	
	5% WB	MCWBR	6.4	7.9	8.7	9.5	8.6	7.7	7.7	7.8	9.0	9.5	7.6	6.4	(37)	
		MCWBR	8.2	10.8	13.9	17.0	17.5	18.0	19.4	19.2	18.9	14.7	10.3	8.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.287	0.289	0.298	0.323	0.333	0.333	0.329	0.329	0.320	0.309	0.297	0.283	(40)	
		taud	2.462	2.523	2.552	2.462	2.462	2.482	2.508	2.502	2.519	2.541	2.522	2.475	(41)	
	Ebn at Noon		813	895	936	934	931	929	929	917	897	856	796	775	(42)	
		Edh at Noon	65	76	86	103	107	105	101	98	88	74	62	58	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.15	2.10	3.20	4.73	5.79	6.58	7.31	6.17	4.49	2.64	1.37	0.92	(44)	
	RadStd		0.15	0.25	0.29	0.36	0.43	0.43	0.40	0.36	0.30	0.29	0.16	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-272	N/A	N/A
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+99	+35

Nomenclature: See separate page